

Started on Tuesday, 1 February 2022, 9:25 AM

State Finished

Completed on Tuesday, 1 February 2022, 10:54 AM

Time taken 1 hour 28 mins

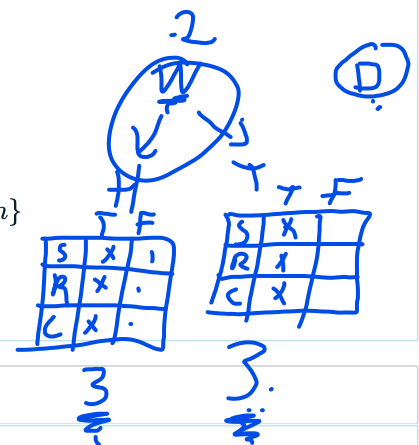
Marks 61.00/70.00

Grade 87.14 out of 100.00

Information

Consider a world defined by the following set $\mathcal{X}$ of random variables:

$Weather(W) = \{sunny, rainy, cloudy\}$
 $Humidity(H) = \{low, high\}$
 $Temperature(T) = \{low, high\}$
 $DayOfWeek(D) = \{mon, tue, wed, thu, fri, sat, sun\}$



Information

Think of the following distributions (or sets of distributions) as multi-dimensional tables. For each of the following items, say how many dimensions (axes) the corresponding table has, and how many entries (numbers) this table will contain (do not worry about redundancies):

Question 1

Correct

Mark 2.00 out of 2.00

$$P(D, W \mid H, T)$$

dims = # of variables not defined to a specific value

Dimensions:



Entries:



entries = # of values of each variable multiplied

Question 2

Correct

Mark 2.00 out of 2.00

$$P(W, H, T = high, D = wed)$$

Dimensions:



Entries:



Question 3

Correct

Mark 2.00 out of 2.00

$$P(T, H \mid D = sat)$$

Dimensions:



Entries:

**Question 4**

Correct

Mark 2.00 out of 2.00

$$P(W \mid H = low, T = high)$$

Dimensions:



Entries:



Question 5

Correct

Mark 2.00 out of 2.00

If we know that $W, H, T \perp D$ (that is, DayOfWeek and Weather, Humidity, Temperature are independent), then how many non-redundant parameters do we need to represent the full joint distribution?

Answer:

The correct answer is:

Handwritten notes for Question 5:

- Diagram showing independence: $W, H, T \perp D$. Arrows point from D to W, H, T .
- Equation: $3 \times 3 \times 2 = 18$ (for W, H, T) and 2 (for D).
- Equation: $18 - 1 = 17$ (non-redundant parameters).
- Diagram showing a 3x3x2 grid for W, H, T and a 2x2 grid for D .
- Equation: $3 \times 3 \times 2 = 18 - 1 = 17$.

Question 6

Correct

Mark 2.00 out of 2.00

How many non-redundant parameters would need to be specified for the following simple Bayesian Network: no - 1

Answer:

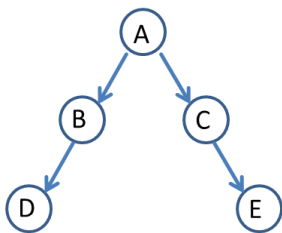
The correct answer is:

Handwritten notes for Question 6:

- Diagram showing a simple Bayesian Network: $D \rightarrow W \rightarrow T \rightarrow H$.
- Equation: $6 + 14 + 3 + 2 = 25$ (total parameters).
- Equation: $25 - 1 = 24$ (non-redundant parameters).
- Diagram showing a 3x3x2 grid for W, H, T and a 2x2 grid for D .
- Equation: $3 \times 3 \times 2 = 18 - 1 = 17$ (non-redundant parameters for W, H, T).
- Equation: 2 (non-redundant parameters for D).

Information

Consider the following Bayesian network over 5 Boolean random variables A, B, C, D, E :



Information

Which of the following statements about this BN are correct?

Question 7

Correct

Mark 1.00 out of 1.00

If A is Boolean with $P(a) = P(\neg a) = 0.5$, then $P(D \mid a) = P(D \mid \neg a)$

Select one:



The correct answer is 'True'



Question 8

Correct

Mark 1.00 out of 1.00

If we removed the link from C to E , the result would not be a valid Bayes Net structure any more

Select one:



The correct answer is 'True'



Question 9

Correct

Mark 1.00 out of 1.00

$$P(A) = P(A|E)$$

Select one:



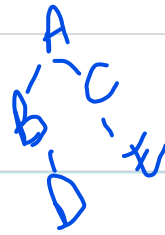
The correct answer is 'True'



Question 10

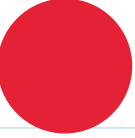
Correct

Mark 1.00 out of 1.00



$$P(A, B, D, C, E) = P(C, E|A)P(A)P(B, D|A)$$

Select one:



The correct answer is

**Question 11**

Correct

Mark 1.00 out of 1.00

If we reversed the direction of all arrows, the resulting network would represent the same set of independencies

Select one:



The correct answer is

**Question 12**

Incorrect

Mark 0.00 out of 1.00

$$P(B|C) = \frac{P(B, C)}{P(C)}$$

$$P(B, C) = P(B|C)P(C)$$

Select one:



$$P(B|C) = P(B, C)/P(C)$$

The correct answer is



Question 13

Correct

Mark 1.00 out of 1.00

$$P(A, B, D, C, E) = P(B, D)P(A)P(C, E)$$

Select one:



The correct answer is



A

B ⊥ D | A

B C

D E

Question 14

Correct

Mark 1.00 out of 1.00

Is B conditionally independent of D , if we know the value of A ?

Select one:



A, B conditionally independent given C if
 $P(A \cap B | C) = P(A | C) P(B | C)$

The correct answer is

**Question 15**

Correct

Mark 1.00 out of 1.00

How many pairs of variables X, Y are there in the network that are (marginally) independent:

$$X \perp Y ?$$

$$P(X|Y) = P(X)$$

Answer:





The correct answer is:



Information

For each of the following queries, say if answering it requires:

- Causal Reasoning
- Evidential Reasoning
- Intercausal Reasoning
- a mixture of several of these
- none of these
- is not a permitted query

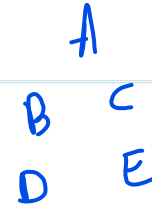
dims = # of

Question 16

Correct

Mark 1.00 out of 1.00

$P(D, C, E) = ?$



- ☐ Causal Reasoning
- ☐ Evidential Reasoning
- ☐ Intercausal Reasoning
- ☐ a mixture of several of these
- ☒ none of these
- ☐ is not a permitted query

The correct answer is:

Question 17

Correct

Mark 1.00 out of 1.00

$P(A \mid D, B) = ?$

- ☐ Causal Reasoning
- ☐ Evidential Reasoning
- ☐ Intercausal Reasoning
- ☐ a mixture of several of these
- ☒ none of these
- ☐ is not a permitted query

The correct answer is:

Question 18

Correct

Mark 1.00 out of 1.00

$P(A) = ?$

dims = # o

- ☒ Causal Reasoning
- ☒ Evidential Reasoning
- ☒ Intercausal Reasoning
- ☒ a mixture of several of these
- ☒ none of these
- ☒ is not a permitted query

Your answer is correct.

The correct answer is:

Causal Reasoning, Evidential Reasoning, Intercausal Reasoning, a mixture of several of these

Question 19

Incorrect

Mark 0.00 out of 1.00

$P(B | C) = ?$

- ☒ Causal Reasoning
- ☒ Evidential Reasoning
- ☒ Intercausal Reasoning
- ☒ reasoning between the causes with a common effect,
- ☒ a mixture of several of these
- ☒ none of these
- ☒ is not a permitted query

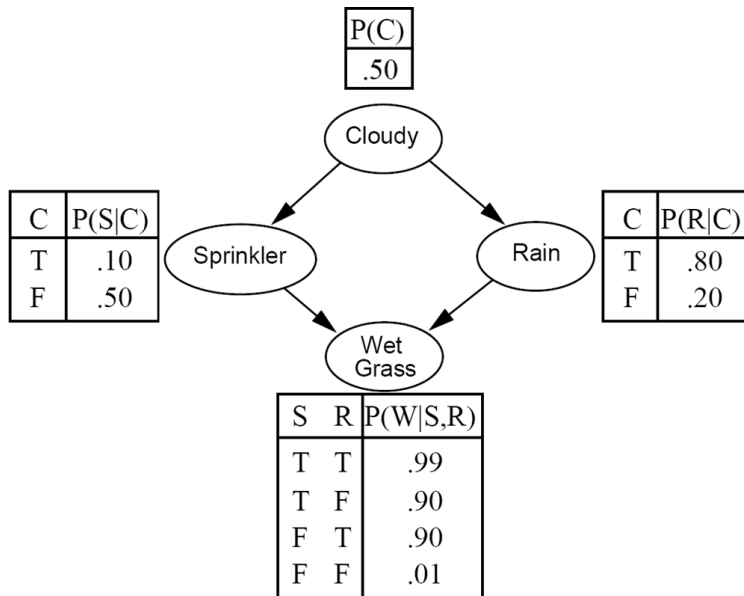
The correct answers are:

Intercausal Reasoning,

reasoning between the causes with a common effect,

Information

Consider the following Bayesian network that models the relation between weather, the state of the lawn in front of your house, and an automatic sprinkler you have installed there, with four Boolean variables abbreviated as C , S , R , W .



Consider the probabilistic query $P(S, R \mid W = \text{true}) = P(S, R \mid w) = ?$.

Question 20

Correct

Mark 2.00 out of 2.00

To answer the query with exact inference, how many different atomic events would Inference by Enumeration need to sum over (total over all combinations for the query variables)?

Answer:



The correct answer is: 8

Question 21

Incorrect

Mark 0.00 out of 2.00

What is the weight that Likelihood Weighting would assign to the sample $(\neg c, \neg s, r, w)$?

Answer:



there is no evidence given except for $W=\text{true}$ so you have only it's weight

The correct answer is: 0.01

Question 22

Incorrect

Mark 0.00 out of 2.00

If we tried to answer this query by **Rejection Sampling**, what distribution would we have to sample variable C from?

$P(C, w)$ ●

$P(C \mid S, R)$ ●

$P(C)$ ●

$P(C \mid S, R, w)$ ●

$P(C \mid w)$ ●



Your answer is incorrect.

The correct answer is:



Question 23

Correct

Mark 2.00 out of 2.00

How many edges (links) would the **Mutilated Network** have that corresponds to this query?

Answer:



The correct answer is:



Question 24

Correct

Mark 2.00 out of 2.00

Which of the following expressions correctly describes the **Resampling Distribution** for variable $Rain(R)$ in **Gibbs Sampling**?

☒ $1/Z \times P(R | C)P(R)$

☐ $1/Z \times P(R)$

☐ $1/Z \times P(R | C)P(W | R, S)$

☐ $1/Z \times P(W | R)P(R | C)P(C)$

☐ $1/Z \times P(R, S, C, w)$

☐ $1/Z \times P(R)P(W | R)$

☐ $1/Z \times P(R, S, W)$

☐ $1/Z \times P(R | w)$

Your answer is correct.

The correct answer is:

Information

Which of the following statements are true: There can be **no** model $\mathcal{M}$ and dataset $\mathcal{D}$ (with discrete variables only) such that the likelihood ...

(Beware of the negation: We are asking whether it is impossible that the likelihood ...)

Question 25

Correct

Mark 1.00 out of 1.00

$L(\mathcal{M} : \mathcal{D}) = 1$

Select one:



The correct answer is '1'

Question **26**

Correct

Mark 1.00 out of 1.00

$$L(\mathcal{M} : \mathcal{D}) < 1$$

Select one:



The correct answer is 'True'



Question **27**

Correct

Mark 1.00 out of 1.00

$$L(\mathcal{M} : \mathcal{D}) = 0$$

Select one:



The correct answer is 'False'



Question **28**

Correct

Mark 1.00 out of 1.00

$$L(\mathcal{M} : \mathcal{D}) = \log L(\mathcal{M} : \mathcal{D})$$

Select one:



The correct answer is 'True'



Information

If we created a new dataset $\mathcal{D}2$ from a dataset $\mathcal{D}$ by duplicating each individual example in $\mathcal{D}$ (so that every example from $\mathcal{D}$ appears twice in $\mathcal{D}2$), which of the following sentences are true, if $0 < L(\mathcal{M} : \mathcal{D}) < 1$:

Question **29**

Correct

Mark 1.00 out of 1.00

For any fixed model $\mathcal{M}$, $L(\mathcal{M} : \mathcal{D}^2) > L(\mathcal{M} : \mathcal{D})$

Select one:



The correct answer is 'True'

Question **30**

Correct

Mark 1.00 out of 1.00

For any fixed model $\mathcal{M}$, $L(\mathcal{M} : \mathcal{D}^2) = 2 * L(\mathcal{M} : \mathcal{D})$

Select one:



The correct answer is 'False'

Question **31**

Correct

Mark 1.00 out of 1.00

For any fixed model $\mathcal{M}$, $L(\mathcal{M} : \mathcal{D}^2) = \sqrt{L(\mathcal{M} : \mathcal{D})}$

Select one:



The correct answer is 'False'

Question 32

Correct

Mark 1.00 out of 1.00

For any fixed model $\mathcal{M}$, $L(\mathcal{M} : \mathcal{D}^2) = L(\mathcal{M} : \mathcal{D})$

Select one:



The correct answer is



Information

Which of the following are true:

Bayesian parameter estimation ...

Question 33

Correct

Mark 1.00 out of 1.00

is based on Bayes' rule

Select one:



The correct answer is



Question 34

Correct

Mark 1.00 out of 1.00

returns zero for any parameter value whose prior probability is zero

Select one:



The correct answer is



Question 35

Correct

Mark 1.00 out of 1.00

cannot lead to the exact same optimal parameter estimate as the Maximum Likelihood Estimate

Select one:



The correct answer is



Question 36

Correct

Mark 1.00 out of 1.00

degenerates to Maximum Likelihood Estimation when run with a uniform prior distribution and posterior mean selection

Select one:



The correct answer is



Information

Consider a world with two Boolean variables X, Y with full joint distribution

$P(X, Y)$

	y^0	y^1
x^0	.25	.25
x^1	.25	.25

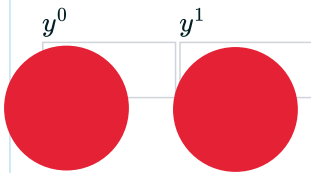
Question 37

Correct

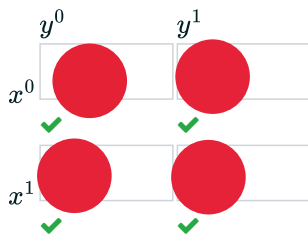
Mark 2.00 out of 2.00

Given a network with structure $Y \longrightarrow X$, construct conditional probability tables for this network such that the resulting Bayes net represents the above full joint distribution.

$$P(Y)$$



$$P(X | Y)$$

**Information**

Which of the following Bayes net structures can correctly model this joint distribution (with appropriate CPD table)?

Question 38

Correct

Mark 1.00 out of 1.00

$$X \longleftarrow Y$$

Select one:



The correct answer is



Question **39**

Correct

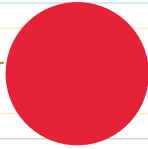
Mark 1.00 out of 1.00

$$X \longrightarrow Y$$

Select one:



The correct answer is



Question **40**

Correct

Mark 1.00 out of 1.00

$$X \longleftrightarrow Y$$

Select one:



The correct answer is



Question **41**

Correct

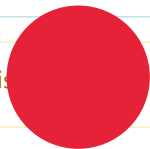
Mark 1.00 out of 1.00

$$X \ Y$$

Select one:



The correct answer is



Question 42

Correct

Mark 1.00 out of 1.00

$Y \rightarrow X$

Select one:

The correct answer is

Information

Suppose we have a training set $\mathcal{D}$ of 100 examples sampled i.i.d. from the above joint distribution. Which of the following statements are true (always assuming a network graph is parametrised with its ML parameters on $\mathcal{D}$)?

Question 43

Correct

Mark 1.00 out of 1.00

If we replaced the minus ($-$) with a plus ($+$) in the BIC score, then a learning algorithm using this score would always learn the model $X \rightarrow Y$, regardless of the data in $\mathcal{D}$

Select one:

The correct answer is

Question 44

Correct

Mark 1.00 out of 1.00

The BIC score will prefer structure $X \rightarrow Y$ to structure $X \leftarrow Y$, because it requires fewer parameters

Select one:

The correct answer is

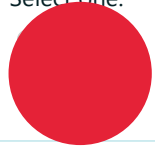
Question 45

Correct

Mark 1.00 out of 1.00

Model structure $X \perp Y$ can never have a higher likelihood on $\mathcal{D}$ than model $X \rightarrow Y$

Select one:



The correct answer is



Question 46

Correct

Mark 1.00 out of 1.00

The BIC score will always prefer structure $X \perp Y$ to structure $X \leftarrow Y$, because it requires strictly fewer parameters

Select one:



The correct answer is



Question 47

Correct

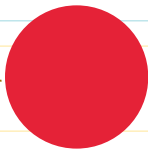
Mark 1.00 out of 1.00

Model structure $X \rightarrow Y$ can never have a higher likelihood on $\mathcal{D}$ than model $X \perp Y$

Select one:

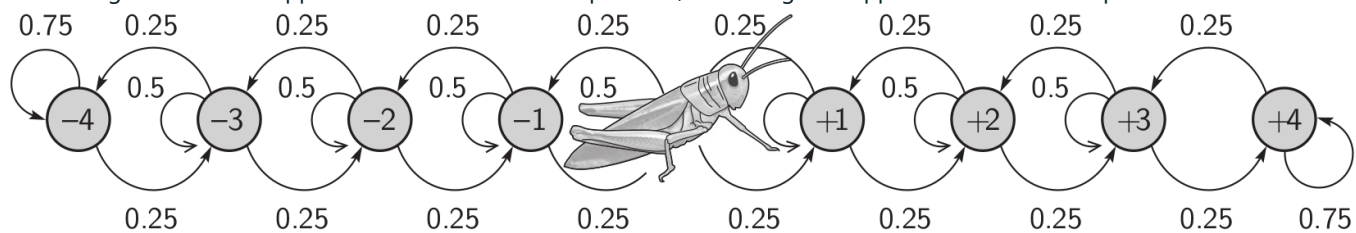


The correct answer



Information

Consider again the Grasshopper Markov Chain from Chapter 04b, with the grasshopper known to start in position 0:



Question 48

Correct

Mark 3.00 out of 3.00

Let us write down the Grasshopper Markov Chain in formal terms. We define the state variable

$$S = \{-4, -3, -2, -1, 0, +1, +2, +3, +4\}.$$

1. Fill in the values for the initial state distribution model:

$$\boldsymbol{\Pi}^T = [$$

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓]

2. Fill in rows 1 and 2 of the state transition model:

$$\mathbf{A}[1, :] = P(S' \mid S = -4) =$$

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓]
 $\mathbf{A}[2, :] = P(S' \mid S = -3) =$

[

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓ ,

✓]

Question 49

Correct

Mark 2.00 out of 2.00

Let's turn this into a Hidden Markov Model. Assume we cannot see the grasshopper, but it makes a *noise* that we can hear from a certain direction:

$$O = \{n_{-4}, n_{-3}, n_{-2}, n_{-1}, n_0, n_{+1}, n_{+2}, n_{+3}, n_{+4}\}$$

where, e.g., n_{-3} means a sound coming from position (state) -3 , and n_0 is a sound coming directly from center state 0 . Let us specify the observation model $\mathbf{B}$ for a world where the sound produced corresponds to the true position of the grasshopper, but we as observers have a slight hearing imprecision (a bias toward the center): we perceive the sound correctly with 90% probability, in the remaining cases hear it one step closer to the center.

1. Fill in the values for columns 4 and 5 (pertaining to states -1 and 0 , respectively) of the observation model $\mathbf{B}$:

$$\mathbf{B}[:, 4]^T = P(O \mid S = -1) =$$

	
✓,	
	
✓,	
	
✓,	
	
✓,	
	
✓,	
	
✓,	
	
✓,	
	
✓]	

$$\mathbf{B}[:, 5]^T = P(O \mid S = 0) =$$

	
✓,	
	
✓,	
	
✓,	
	

✓ ,
☐

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Information

If we had no hearing imprecision -- that is, if we always correctly heard and recognised a sound n_i when the grasshopper is in position i -- and if we formalised this in a corresponding observation model $\mathbf{B}$, which of the following statements would be true?

Question 50

Incorrect

Mark 0.00 out of 1.50

The likelihood of this model w.r.t. any observation sequence would be 1.0

Select one:

☐

The correct answer is

☐

Question 51

Correct

Mark 1.50 out of 1.50

In the Viterbi algorithm, all back pointers at each time step $t > 1$ will point back to the central state 0 at $t - 1$, for any observation sequence.

Select one:

☐

The correct answer

☐

Question **52**

Incorrect

Mark 0.00 out of 1.50

The probability of every observation sequence under this model would be 1.0

Select one:



The correct answer is



Question **53**

Correct

Mark 1.50 out of 1.50

The forward algorithm can completely ignore the observations.

Select one:



The correct answer is



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